

First Order Circuits

Analytical expression for V_{out} in lab manual figure 4-3

fig 4-3 (a)

$$\begin{aligned}V_{\text{out}} &= V_{C(t)} = E\left(1 - e^{-\frac{t}{RC}}\right) \\&= E\left(1 - e^{-\frac{t}{1k\Omega \times 0.1\mu F}}\right) \\V_{\text{out}(t)} &= E\left(1 - e^{-\frac{t}{100\mu s}}\right)\end{aligned}$$

```
fplot(@(x) 1-exp(-x/100),  
[0,500])  
xlabel('t','FontSize',14)  
ylabel('E','FontSize',14)  
set(gca,'FontSize',14)  
grid on  
print -dsvg fig4-3a.svg
```

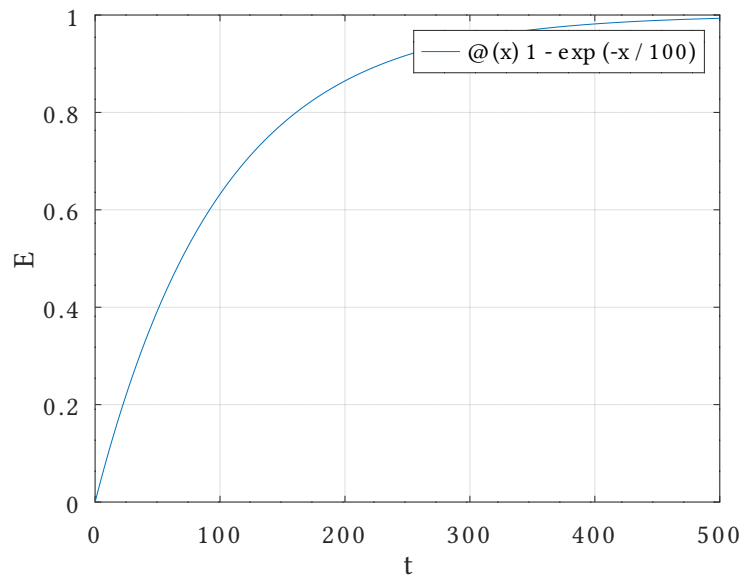


fig 4-3 (b)

$$\begin{aligned}V_{\text{out}} &= V_{R(t)} = Ee^{-\frac{t}{RC}} \\&= Ee^{-\frac{t}{1k\Omega \times 0.1\mu F}} \\V_{\text{out}(t)} &= Ee^{-\frac{t}{100\mu s}}\end{aligned}$$

```
fplot(@(x) exp(-x/100),
[0,500])
xlabel('t','FontSize',14)
ylabel('E','FontSize',14)
set(gca,'FontSize',14)
grid on
print -dsvg fig4-3b.svg
```

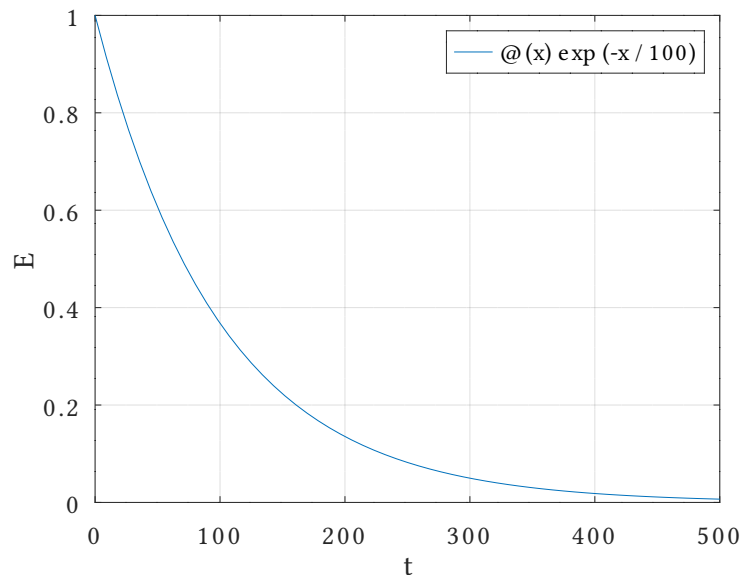


fig 4-3 (c)

$$V_{\text{out}} = V_{L(t)} = E e^{-\frac{t}{\tau}}$$

$$= E e^{-\frac{100 \text{ mH}}{1 \text{ k}\Omega} t}$$

$$V_{\text{out}} = E e^{-\frac{t}{100 \mu\text{s}}}$$

```
fplot(@(x) exp(-x/100),
[0,500])
xlabel('t','FontSize',14)
ylabel('E','FontSize',14)
set(gca,'FontSize',14)
grid on
print -dsvg fig4-3c.svg
```

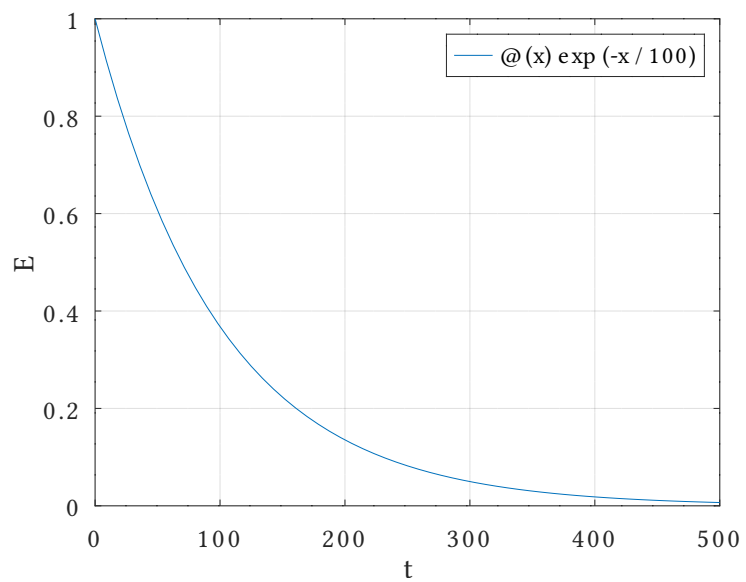


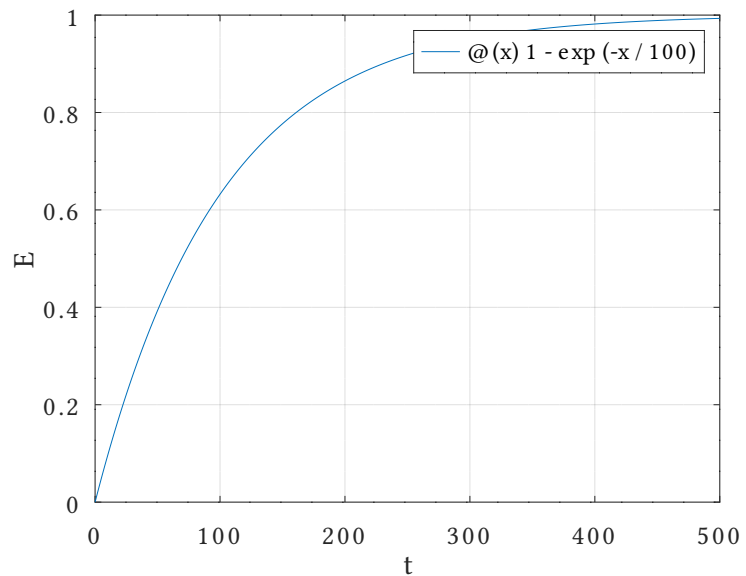
fig 4-3 (d)

$$V_{\text{out}} = V_{R(t)} = E \left(1 - e^{-\frac{t}{R}} \right)$$

$$= E e^{-\frac{100mH}{1k\Omega} \frac{t}{H}}$$

$$V_{\text{out}(t)} = E e^{-\frac{t}{100\mu s}}$$

```
fplot(@(x) 1-exp(-x/100),
[0,500])
xlabel('t','FontSize',14)
ylabel('E','FontSize',14)
set(gca,'FontSize',14)
grid on
print -dsvg fig4-3d.svg
```



Square wave

fig 4-4 (a)

$$V_{\text{out}} = \Sigma V_C = V_{C(0s < t < 10s)} + V_{C(10s < t < 20s)} + V_{C(20s < t < 30s)}$$

$$= E \left(1 - e^{-\frac{t}{100\mu s}} \right) - 2E \left(1 - e^{-\frac{t-5}{100\mu s}} \right) + 3E \left(1 - e^{-\frac{t-10}{100\mu s}} \right) + \dots$$

fig 4-4 (b)

$$V_{\text{out}} = V_{R(t)} = E e^{-\frac{t}{RC}}$$

$$= E e^{-\frac{t}{1k\Omega \times 0.1\mu F}}$$

$$V_{\text{out}(t)} = E e^{-\frac{t}{100\mu s}}$$

fig 4-4 (c)

$$V_{\text{out}} = V_{L(t)} = E e^{-\frac{t}{L}})$$

$$= E e^{-\frac{100mH}{1k\Omega} \frac{t}{H}}$$

$$V_{\text{out}} = E e^{-\frac{t}{100\mu s}}$$

fig 4-4 (d)

$$\begin{aligned}V_{\text{out}} &= V_{R(t)} = E \left(1 - e^{-\left(\frac{t}{\tau}\right)} \right) \\&= E e^{-\frac{t}{\frac{100mH}{1k\Omega}}} \\V_{\text{out}(t)} &= E e^{-\frac{t}{100\mu s}}\end{aligned}$$

Second Order Circuits

Fig 4-5

Fig 4-5 (a)

$$\begin{aligned}s^2 + \frac{R}{L}s + \frac{1}{LC} &= 0 \\ \alpha &= \frac{R}{2L} = \frac{R}{2 \times 100mH} = 5R & \alpha^2 &= 25R^2 \\ \omega_0^2 &= \frac{1}{LC} = \frac{1}{100mH \times 0.01\mu F} = \frac{1}{10^{-9}} = 1.0 \times 10^9\end{aligned}$$

Over-damped response

$$\begin{aligned}\alpha^2 &> \omega_0^2 \\ 25R^2 &> 1 \times 10^9 \\ R^2 &> 4 \times 10^7 \\ R &< 6.32k\Omega\end{aligned}$$

When $R < 6.32k\Omega$, the circuit is Over-damped

Critically-damped response

$$\begin{aligned}\alpha^2 &= \omega_0^2 \\ R^2 &= 4 \times 10^7 \\ R &= 6.32k\Omega\end{aligned}$$

When $R = 6.32k\Omega$, the circuit is Critically-damped

Under-damped response

$$\begin{aligned}\alpha^2 &< \omega_0^2 \\ R^2 &< 4 \times 10^7 \\ R &> 6.32k\Omega\end{aligned}$$

When $R > 6.32k\Omega$, the circuit is Under-damped

Fig 4-5 (b)

Characteristic equation:

$$s^2 + \frac{1}{RC}s + \frac{1}{LC} = 0$$
$$\alpha = \frac{1}{2RC} = \frac{1}{2 \times R \times 0.01\mu F} = \frac{5 \times 10^7}{R}$$
$$\omega_0^2 = \frac{1}{LC} = \frac{1}{100mH \times 0.01\mu F} = \frac{1}{10^{-9}} = 1.0 \times 10^9$$

Over-damped response

$$\alpha^2 > \omega_0^2$$
$$\frac{25 \times 10^{14}}{R^2} > 1 \times 10^9$$
$$R < 1.58k\Omega$$

When $R < 1.58k\Omega$, the circuit is Over-damped

Critically-damped response

$$\alpha^2 = \omega_0^2$$
$$R = 1.58k\Omega$$

When $R = 1.58k\Omega$, the circuit is Critically-damped

Under-damped response

$$\alpha^2 < \omega_0^2$$
$$R^2 < 4 \times 10^7$$
$$R > 1.58k\Omega$$

When $R > 1.58k\Omega$, the circuit is Under-damped